

B Academy
RO

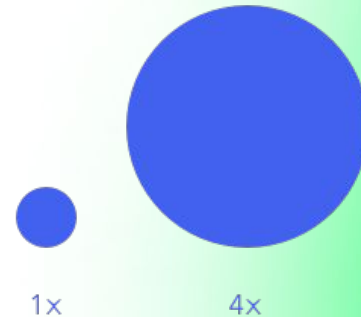
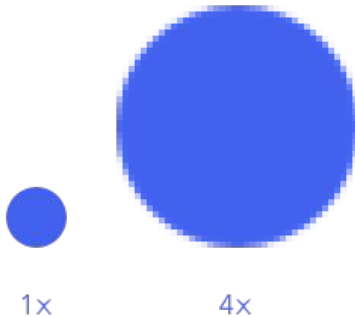


Manual

SVG

VS

Raster	Vector
Consist of pixels	Consist of mathematical formulas, shapes, and lines
Large file size	Very small file size
Scales with quality loss	Looks the same at any size
Can only be converted to other raster formats	Can be converted to raster images
Cannot be modified easily	Can be easily modified (e.g., splitting into components, animating)
Ideal for photographs	Ideal for icons
Uses RGB color space	



Vector images

Scalable Vector Graphics



A markup language extended from XML for describing two-dimensional vector graphics

SVG files can be created and edited with any text editor and drawing programs

```
.block {  
    background-image: url(image.svg);  
    list-style-image: url(image.svg);  
    content: url(image.svg);  
}  

```

```
<svg viewBox="0 0 30 10" width="300"  
height="200">  
    <rect width="100%" height="100%" fill="red"  
/>  
</svg>  
  
<svg viewBox="0 0 30 10">  
    <use xlink:href="image.svg" />  
</svg>
```



Common Attributes

width

height

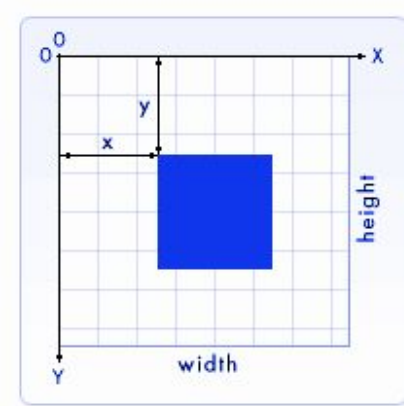
x – Horizontal coordinate

y – Vertical coordinate

fill – Fill color

stroke – Stroke color

stroke-width



- Clipping, Masking and Compositing properties:

- ['clip-path'](#)
- ['clip-rule'](#)
- ['mask'](#)
- ['opacity'](#)
- ['marker-end'](#)
- ['marker-mid'](#)
- ['marker-start'](#)
- ['shape-rendering'](#)

- Filter Effects properties:

- ['enable-background'](#)
- ['filter'](#)
- ['flood-color'](#)
- ['flood-opacity'](#)
- ['lighting-color'](#)
- ['stroke'](#)
- ['stroke-dasharray'](#)
- ['stroke-dashoffset'](#)
- ['stroke-linecap'](#)
- ['stroke-linejoin'](#)
- ['stroke-miterlimit'](#)

- Gradient properties:

- ['stop-color'](#)
- ['stop-opacity'](#)
- ['stroke-opacity'](#)
- ['stroke-width'](#)
- ['text-rendering'](#)

- Interactivity properties:

- ['pointer-events'](#)

- Color and Painting properties:

- ['color-interpolation'](#)
- ['color-rendering'](#)
- ['fill'](#)
- ['fill-opacity'](#)
- ['fill-rule'](#)
- ['image-rendering'](#)
- ['marker'](#)

- Text properties:

- ['alignment-baseline'](#)
- ['baseline-shift'](#)
- ['dominant-baseline'](#)
- ['glyph-orientation-horizontal'](#)
- ['glyph-orientation-vertical'](#)
- ['text-anchor'](#)
- ['writing-mode'](#)

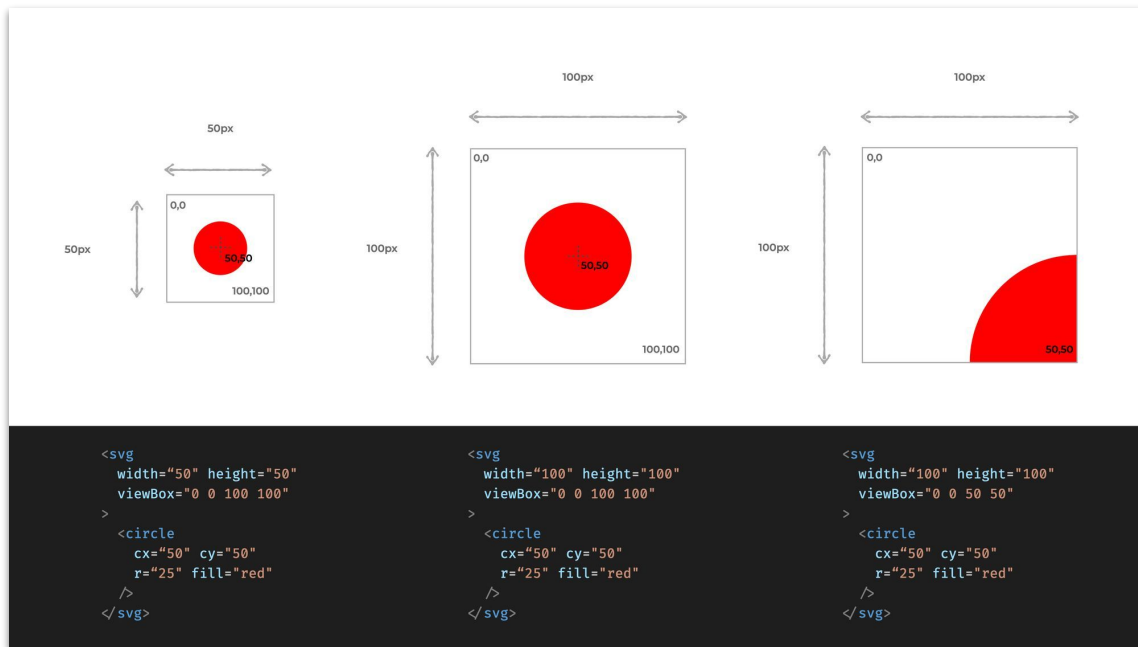


viewBox

- Defines the aspect ratio
- Responsible for scaling

viewBox="0 0 24 24"

- coordinate x
- coordinate y
- width
- height



Anatomy of

SVG

<circle>

<rect>

<path>

<ellipse>

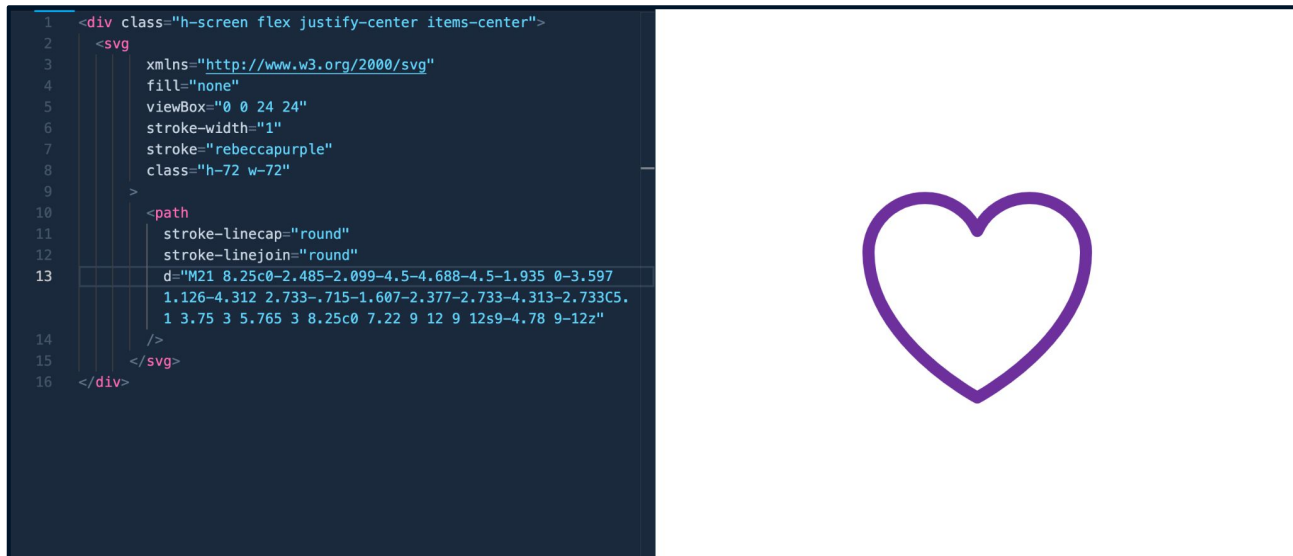
<polygon>

<text>

Grouping container – <g>

Hidden element – <defs>

Reuse an element – <use>



Layering of shapes goes from top to bottom in the code; the last element is most visible.



rect

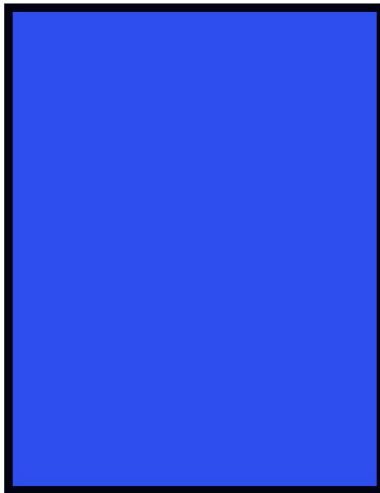
width

height

fill

stroke

stroke-width



```
...  
<rect width="180"  
height="220"  
style="fill:pink;  
stroke-width:4;  
stroke:rgb(0,0,0)">  
...
```



circle

fill

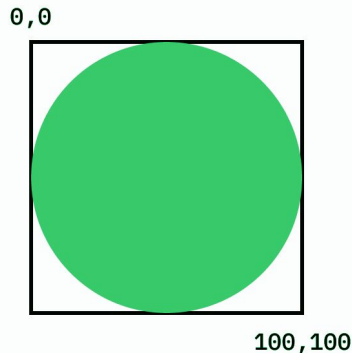
stroke

stroke-width

cx – central horizontal
coordinate

cy – central vertical
coordinate

r – radius



```
<svg width="100"  
height="100">  
  <circle cx="50"  
    cy="50" r="50"  
    fill="yellow" />  
</svg>
```



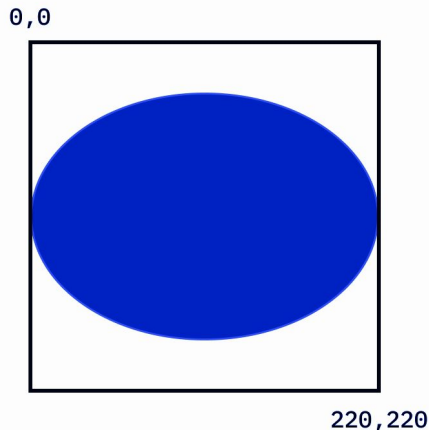

ellipse

cx – central horizontal coordinate

cy – central vertical coordinate

rx – horizontal radius

ry – vertical radius



```
...  
<ellipse  
  cx="110" cy="110"  
  rx="100" ry="70"  
  style="fill:purple;  
        stroke:pink">  
...
```



text

x

y

fill

A diagram showing a white rectangular box with a thin black border. Inside the box, the word 'HTML' is written in a bold, blue, sans-serif font, centered horizontally and vertically.

HTML

```
...<text x="50" y="50"  
fill="pink">HTML</text>...
```



line

x1/x2 – first/last
horizontal coordinate

y1/y2 – first/last vertical
coordinate



```
...<line x1="0" y1="0" x2="120"  
y2="0" style="stroke:blue" />...
```



```
...<line x1="0" y1="0" x2="120" y2="0"  
style="stroke-dasharray:10,10" />...
```



```
...<line x1="0" y1="0" x2="120"  
y2="0" style="stroke-width:6" />...
```



polygon

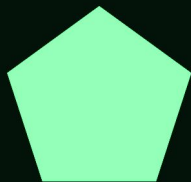
points – vertex

coordinates

fill

stroke

stroke-width



```
<svg...>
```



```
points="..."
```

```
style="..." />
```

```
</svg>
```



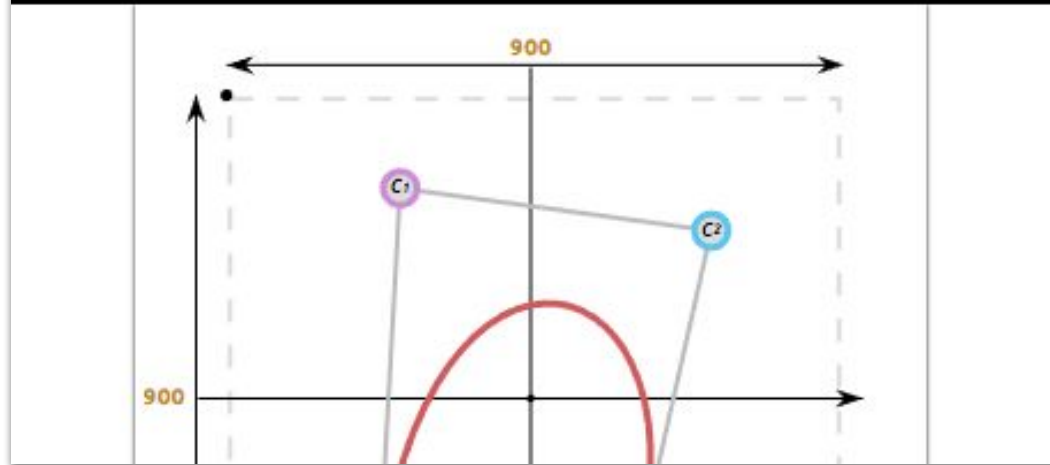
path

line for drawing figures

d – listing of coordinates



```
<svg viewBox=' -450 -450 900 900 '>  
  <path d='M -229,333  
          C -193,-312 267,-249 161,222 '/>  
</svg>
```



base64

✗ don't use

base64 at all —
with raster images —

```

```

```

```

Sprite

use sprites to join all images
network requests

can be done with generators
like [spritecow.com](https://www.spritecow.com)

to not do 1x and 2x sprites
can be used SVG



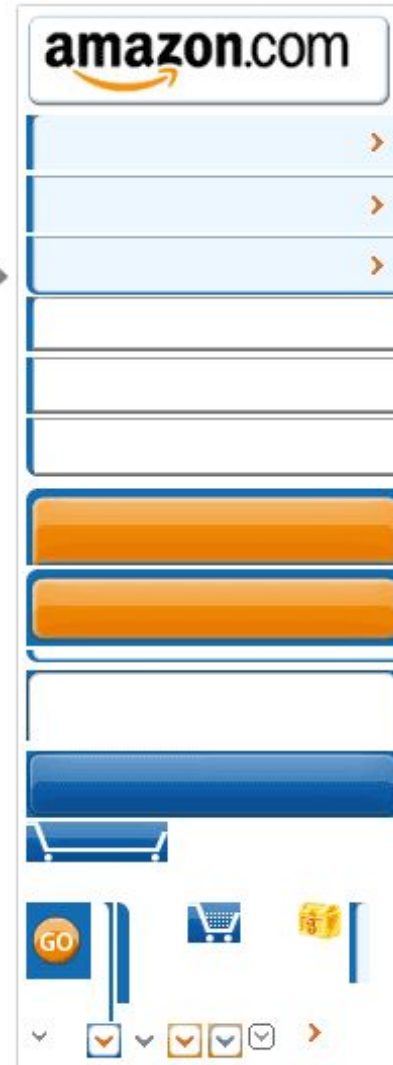
← Google



← YouTube



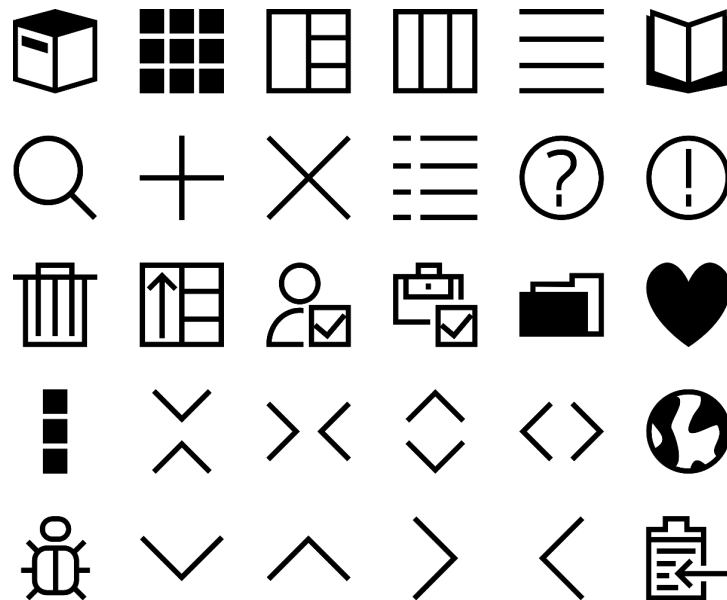
← Facebook



Amazon →

SVG Sprites

- Vector graphics (sharp at any size)
- One HTTP request
- Style with CSS (color, size, animations!)
- Easily accessible
- Light and fast!

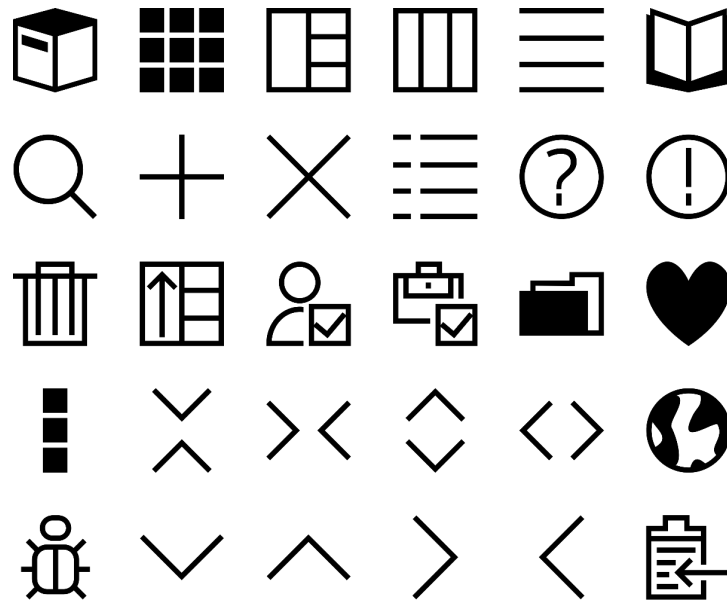


How SVG Sprites Work

SVG sprite = one file with many symbols.

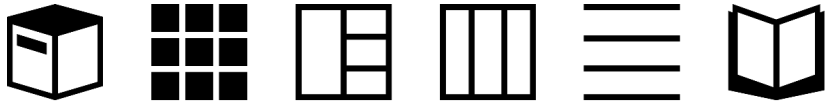
Each icon is a **<symbol>** with an id.

Use **<use>** to reference it anywhere on the page.



```
<svg viewBox="0 0 100 100" fill="none"
xmlns="http://www.w3.org/2000/svg">
```

```
<symbol id="box">
  // ...
</symbol>
```



```
<symbol id="search">
  // ...
</symbol>
```

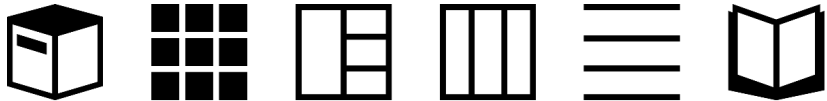


```
</svg>
```

```
<svg class="icon">
  <use xlink:href="./sprite.svg#box"></use>
</svg>
```

```
<svg viewBox="0 0 100 100" fill="none"
xmlns="http://www.w3.org/2000/svg">
```

```
<symbol id="box">
  // ...
</symbol>
```



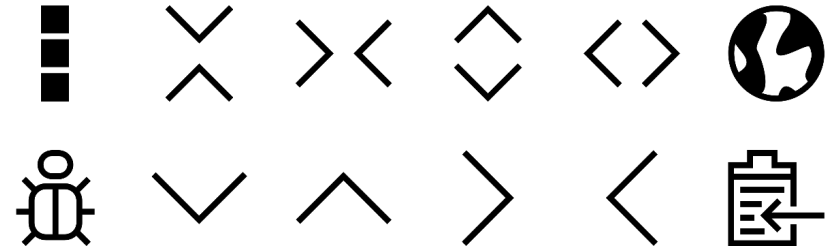
```
<symbol id="search">
  // ...
</symbol>
```



```
<symbol id="bin">
  // ...
</symbol>
```



```
<symbol id="bar">
  // ...
</symbol>
```



```
</svg>
```

Change icon colors on hover

[example →](#)

```
.icon {  
    fill: blue;  
    transition: fill 0.3s, stroke 0.3s;  
}
```

```
.icon:hover {  
    fill: red;  
}
```

color: currentColor

[example](#) →

replace colors inside of SVGs in sprite folder to currentColor

```
<svg viewBox="0 0 512 512" class="logo__icon">  
  <path fill="currentColor" d="m48 300 180 169a41 41  
0 0 0 56 0l180-169c31-28 48-68 48-109v-6A143 143 0 0 0  
268 84l-12 12-12-12A143 143 0 0 0 0 185v6c0 41 17 81 48  
109z" />  
</svg>
```

Custom properties

[example](#) →

replace colors inside of SVGs in sprite folder to custom properties with fallback color

```
<svg viewBox="0 0 512 512" class="logo__icon">  
  <path fill="var(--icon-color, blue)" d="m48 300 180  
169a41 41 0 0 0 56 0l180-169c31-28 48-68 48-109v-6A143 143  
0 0 0 268 84l-12 12-12-12A143 143 0 0 0 0 185v6c0 41 17 81  
48 109z" />  
</svg>
```

Custom properties +currentColor

replace colors inside of SVGs in sprite folder to custom properties with fallback color as a currentColor

```
<svg viewBox="0 0 512 512" class="logo__icon">
  <path fill="var(--icon-color, currentColor)" d="m48
300 180 169a41 41 0 0 0 56 0l180-169c31-28 48-68
48-109v-6A143 143 0 0 0 268 84l-12 12-12-12A143 143 0 0 0
0 185v6c0 41 17 81 48 109z" />
</svg>
```